TITLE PLACEHOLDER

AUTHORS PLACEHOLDER

Preprint draft, November 8, 2026

Abstract

ABSTRACT PLACEHOLDER (written after confirmation results).

1 Introduction

Learning a task that spans hundreds of milliseconds requires assigning credit to synapses whose influence on behaviour is delayed and passes through recurrent dynamics. The exact solution, back-propagation through time (BPTT; Werbos, 1990), needs information that a synapse does not have: the future error and its path through every other neuron. Local alternatives keep, at each synapse, an *eligibility trace* of how the synapse could have changed its postsynaptic cell. They combine it with a broadcast error or reward signal, in line with three-factor plasticity (Gerstner et al., 2018). Examples are RFLO (Murray, 2019) and e-prop (Bellec et al., 2020) (see also Williams and Zipser, 1989; Marschall et al., 2020). These rules are exact for a cell's own dynamics, and they approximate the rest of the network, for example with random feedback (Lillicrap et al., 2016).

What the trace must contain depends on the cell's own dynamics, and real cortical neurons are not the leaky units used in most such work. Their responses depend on several timescales, and most show spike-frequency adaptation (Benda and Herz, 2003). For spiking neurons with a single adaptive threshold, adaptation is known to support memory over behavioural timescales (Bellec et al., 2018; Salaj et al., 2021), and e-prop includes the adaptation variable in its eligibility trace (Bellec et al., 2020). Whether this matters for neurons whose intrinsic dynamics were *fitted to recordings*, with many heterogeneous filters per cell, has been hard to test: data-fitted single-cell models were usually not differentiable (Teeter et al., 2018), and large data-constrained network models have been trained with BPTT (Billeh et al., 2020; Chen et al., 2022). A recent bank of differentiable generalized firing-rate (GFR) models, fitted to more than a thousand patch-clamp recordings (Yeung et al., 2025), now makes the test possible.

We built recurrent networks entirely from these fitted cells, keeping each cell's intrinsic parameters frozen and making only the synaptic weights plastic. We asked three questions: (i) how close a cell-local, forward-in-time rule comes to the exact gradient on working-memory tasks, compared with a frozen recurrent reservoir; (ii) whether the rule must include each cell's fitted spike-history filter in its eligibility trace; and (iii) whether the specific identity of the cells (the natural pairing of each cell's parameters and their heterogeneity) or the type-based wiring matters for learning, once confounds are removed.

The third question carries a trap that we found in our own earlier analyses. Shuffling parameters across cells moves each cell away from the operating point it was calibrated to. In our networks this roughly doubled firing rates and multiplied saturation, so an apparent cost of "destroying biological identity" was in fact a change of operating point. We therefore recalibrate every manipulated bank with the same label-free procedure before comparing. We also split cells by donor animal, compare learners at learning rates tuned on a separate development cohort, treat the network as the unit of inference (Aarts et al., 2014), fit every population-analysis axis on trials independent of those it is evaluated on (Kriegeskorte et al., 2009), and test all hypotheses in a protocol locked before the held-out confirmation cohort was touched.

SUMMARY OF FINDINGS PLACEHOLDER.

2 Results

RESULTS PLACEHOLDER (filled from the confirmation cohort).

3 Discussion

DISCUSSION PLACEHOLDER.

4 Methods

4.1 Data-fitted neuron models

We used the differentiable generalized firing-rate (GFR) models of Yeung et al. (2025). They were fitted to whole-cell current-clamp recordings in the Allen Cell Types Database (Gouwens et al., 2019), and we took the code and data from upstream commit `835a535`. Each cell i has $K = 8$ filters with fixed time constants $\tau_k \in \{\approx 0, 20, 50, 100, 200, 500, 1000, 2000\}$ ms. At the fitted bin size $\Delta t = 20$ ms the decay coefficients are $d_k = 1 - e^{-\Delta t/\tau_k}$, shared by all cells. With input current $I_{i,t}$ (pA) and rate $r_{i,t}$ (Hz),

$$h_{i,t}^{(k)} = (1 - d_k)\, h_{i,t-1}^{(k)} + a_{ik}\, I_{i,t} + \beta_{ik}\, r_{i,t-1}, \tag{1}$$

$$r_{i,t} = \gamma_i \,\mathrm{ReLU}\Big[\tanh\Big(\textstyle\sum_d c_{id}^2\, x_{i,t}^d\Big)\Big], \qquad x_{i,t} = \big(\bar{h}_{i,t} - b_i\big)/\sigma_i, \quad \bar{h}_{i,t} = \tfrac{1}{K}\textstyle\sum_k h_{i,t}^{(k)}. \tag{2}$$

The stored a_{ik} and β_{ik} already include the discretization, so no extra Δt factor is applied. Upstream rates in spikes/ms are converted to Hz once. Fitted parameters are never changed by training. Our implementation reproduces the upstream equation for all 1,124 fitted cells (unit tests).

The coefficients β_{ik} form the cell's spike-history filter. Its steady-state gain $\bar{\beta}_i = \frac{1}{K}\sum_k \beta_{ik}/d_k$ (pA Hz^{-1}) is negative for almost every cell, which is spike-frequency adaptation (Benda and Herz, 2003; Weber and Pillow, 2017): median -10.5 for pyramidal (Pyr) cells, -6.6 for SST, -4.1 for VIP and -0.8 for PV cells in the development cohort.

Cells and cohorts. Following the upstream release criterion, we kept mouse cells with validation explained variance ≥ 0.5. We did not select on the held-out test sweeps. Operational labels came from Cre line and dendrite type: spiny cells from excitatory lines are Pyr, and Pvalb-, Sst- and Vip-Cre cells are PV, SST and VIP. This gave 597 cells (339 Pyr, 107 PV, 86 SST, 65 VIP) from 410 donors, mostly in primary visual cortex. Whole donors were assigned to a development cohort (298 cells) or a confirmation cohort (299 cells). The assignment was a seeded, type-stratified greedy procedure. No donor contributes to both cohorts. All tuning and piloting used only the development cohort.

4.2 Networks

A network has $N = 100$ cells, 80 Pyr, 10 PV, 5 SST and 5 VIP, sampled without replacement for each network seed. The input current to cell i is

$$I_{i,t} = I_i^0 + S_i\Big(\textstyle\sum_c W_{ic}^{\mathrm{in}} u_{c,t} + \sum_j s_j |W_{ij}| M_{ij}\, r_{j,t-1}/R_0\Big), \tag{3}$$

with $R_0 = 100$ Hz for every cell. The sign s_j is $+1$ for Pyr and -1 for interneurons (Dale's law). The mask M is drawn from a type-level connection-probability matrix inspired by cortical motifs (Pfeffer et al., 2013; Pi et al., 2013). These are modelling assumptions, not measured values (Table 1). Magnitudes $|W_{ij}|$ are log-normal with mean 1 ($\sigma_{\log} = 0.35$), scaled by $0.6/\langle\text{in-degree}\rangle$, and inhibitory sources are multiplied by 1.5. Input weights are $W_{ic}^{\mathrm{in}} \sim \mathcal{N}(0, 1/5)$. The readout $z_t = \sum_{i\in\mathrm{Pyr}} w_i r_{i,t}/R_0 + b$ uses Pyr cells only.

Operating-point calibration. For each cell, I_i^0 is chosen on a 33-point grid in $[0, \sigma_i]$ so that its isolated steady-state rate is closest to 5% of its own fitted ceiling γ_i. The rate is measured in the last third of a 3-s constant-current probe. The input scale is $S_i = 0.25\,\sigma_i$. The procedure uses no task information. Because the target is relative to each cell's ceiling, the resulting baseline rates differ by type (PV higher), by construction.

Identity and wiring controls. *Within-type shuffle*: independent permutations, within each type, of the input filters $a_{i\cdot}$, the history filters $\beta_{i\cdot}$ and the activation tuple (σ, γ, c, b). This keeps marginal distributions and breaks the pairing of a cell's own parameters. *Type-homogeneous*: every parameter is replaced by its type mean. *Cross-type permutation*: whole parameter tuples are permuted across types. *Fast decay*: $d_k \leftarrow \max(d_k, 1-e^{-1})$, so $\tau_k \le 20$ ms, with a and β rescaled to keep a/d and β/d, and hence the static response. Unless stated otherwise, each manipulated bank is recalibrated with the same label-free procedure ("matched"). The "reference" calibration keeps the intact bank's I^0, S, as in earlier versions of this project; it is shown as a control. *Degree-preserving rewiring* applies $10|E|$ directed double-edge swaps. It preserves every cell's in- and out-degree, its Dale sign and its multiset of outgoing magnitudes.

4.3 Tasks

Inputs have five channels: two sensory streams (A, B), two cue channels and a go signal. Time is binned at 20 ms: fixation 100 ms, stimulus 200 ms, a silent delay of 100, 200 or 300 ms in training, and a 60-ms response window marked by the go channel. Sensory evidence has signed coherence $\pm\{0.25, 0.5, 1\}$ plus Gaussian noise ($\sigma = 0.25$ per bin). Every batch is exactly balanced over cue, relevant sign and irrelevant sign. *T1-early* (context-dependent decision; Mante et al., 2013): a context cue during fixation and stimulus says which stream is relevant, and the target is the sign of the relevant stream. *T1-late*: the same, but the context cue (60 ms) arrives only after the delay, so both streams must be held. *T2-dms* (delayed match-to-sample, Miller et al., 1996; Masse et al., 2019): stream A is the sample and stream B a simultaneous distractor. A ± 1 probe arrives after the delay, and the target is sign(sample) $\times$ probe, so no input before the probe carries information about the choice. Conflict trials are those whose two streams have opposite signs. Test splits are 512 trials each: in-distribution (ID); unseen delays of 400 and 600 ms (OOD); higher sensory noise ($\sigma = 0.5$); a 40-ms zero-mean distractor ($\sigma = 0.75$) in mid-delay; and all three combined.

4.4 Loss and learning rules

The loss is the binary cross-entropy of z_t at every response bin, averaged within a trial and then over the batch. The choice is the sign of the mean response-window logit. Only $|W|$ (masked, non-negative, at most 2), w and b are plastic.

BPTT. The exact gradient by back-propagation through time (Werbos, 1990) through all filters, the spike-history loop and the recurrent connections. It was checked against central finite differences.

Local rule. The local rule is e-prop (Bellec et al., 2020), which is related to RFLO (Murray, 2019), adapted to GFR cells. It approximates $\partial\mathcal{L}/\partial W_{ij}$ by

$$\hat{g}_{ij} = \sum_t L_{i,t}\,\frac{g'_{i,t}}{R_0}\,\bar{e}_{ij,t}, \qquad L_{i,t} = B_i\,\frac{\partial\mathcal{L}}{\partial z_t}, \tag{4}$$

where B is a fixed random feedback vector (Lillicrap et al., 2016) and g' is the activation slope. The K-dimensional eligibility trace $e_{ij,t} = \partial h_{i,t}/\partial W_{ij}$ follows only cell i's own dynamics, with

presynaptic activity held fixed:

$$e_{ij,t}^{(k)} = (1 - d_k)\, e_{ij,t-1}^{(k)} + a_{ik} S_i s_j \frac{r_{j,t-1}}{R_0} + \underbrace{\beta_{ik}\, g'_{i,t-1}\, \bar{e}_{ij,t-1}}_{\text{spike-history term}}. \tag{5}$$

Every quantity is available at the synapse or the postsynaptic cell, plus one broadcast error signal (Gerstner et al., 2018). *No-β* removes only the spike-history term in Eq. 5; the network dynamics are unchanged. *Symmetric feedback* uses $B = w$ (the readout weights), which isolates the error introduced by ignoring paths through other cells. The quantity in Eq. 5 can be computed forward in time, as a synapse would, or by an equivalent reverse sweep over each cell's own dynamics, which is the network gradient with all paths through other cells removed. The two agree to $< 10^{-10}$ relative error (unit tests). We used the reverse sweep because it is $\sim$17$\times$ faster.

Frozen reservoir. Recurrent weights are fixed, and only w, b are trained, with their exact gradient.

Optimizers. *Adam* (Kingma and Ba, 2015) with global gradient-norm clipping at 1.0 is the primary optimizer. *RMS* is per-parameter RMS normalization without momentum and without clipping (Tieleman and Hinton, 2012), so every update depends only on that synapse's own history. It is an optimizer, not a model of homeostatic synaptic scaling (Turrigiano, 2008). After each update, magnitudes are projected onto $[0, 2]$.

Training and model selection. Each run trains for 3000 updates of 32 trials. Validation uses 256 ID trials every 50 updates, and the checkpoint with the lowest validation BCE (including the untrained network) is kept. Learning rates were chosen on the development cohort only. The grid was $\eta_{\text{rec}} \in \{3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}\} \times \eta_{\text{out}} \in \{0.03, 0.1, 0.3, 1\}$, three seeds per candidate, and the criterion was mean best validation BCE. This was done separately for each learner, task and optimizer (Table **??**). A first search with $\eta_{\text{out}} \leq 0.1$ put every learner at the upper edge of the grid, so the grid was extended upward. No test split had been evaluated at that point. Controls use the rates of their learner. A network seed fixes the sampled cells, graph, initial weights, feedback vector and every trial, so all conditions are paired.

4.5 Analyses

Population codes. For each phase (stimulus, delay, response), we fitted PCA, ridge probes ($\alpha = 1$) for choice, cue and each stream's sign, and a class-mean choice axis on 128 *analysis-fit* trials. All were evaluated on 128 independent *analysis-evaluation* trials, to avoid double dipping (Kriegeskorte et al., 2009).

Gradient alignment. At each saved checkpoint, on 32 independent diagnostic trials, we computed the cosine between the exact recurrent gradient and each local estimate: with or without β, and with random or symmetric feedback. Per-neuron alignment uses the row of the gradient belonging to postsynaptic cell i. Each cell's *adaptation loop gain* is $|\langle g'_i \rangle\, \bar{\beta}_i|$, the steady-state history gain times the mean activation slope on those trials.

4.6 Statistics and protocol

The unit of inference is the network. We avoid pooling cells or trials as if they were independent (Aarts et al., 2014). Comparisons are paired by seed and use exact two-sided sign-flip tests over all 2^{16} sign assignments, with percentile bootstrap CIs over networks. Families of tests and the Holm correction are as preregistered (Section A.1). The 2×2×2 factorial effects (identity, wiring, learner) are main effects and interactions computed within each network. An identity, wiring or heterogeneity effect is called *practically negligible* when its 90% CI lies within ± 5 pp. The

protocol, the analysis code and the source hash were locked before the confirmation cohort was used (Supplementary Section A.1). Results outside it are labelled exploratory.

Code and data availability

All code, configurations, the locked protocol and every run's raw outputs (per-trial predictions, learning curves, checkpoints) are in the project repository. The fitted neuron models and cell metadata are those released by Yeung et al. (2025) (AllenInstitute/GRNN, commit `835a535`) under their original licence.

References

E. Aarts, M. Verhage, J. V. Veenvliet, C. V. Dolan, and S. van der Sluis. A solution to dependency: using multilevel analysis to accommodate nested data. *Nature Neuroscience*, 17:491–496, 2014. doi: 10.1038/nn.3648.

G. Bellec, D. Salaj, A. Subramoney, R. Legenstein, and W. Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems 31*, 2018.

G. Bellec, F. Scherr, A. Subramoney, E. Hajek, D. Salaj, R. Legenstein, and W. Maass. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11: 3625, 2020. doi: 10.1038/s41467-020-17236-y.

J. Benda and A. V. M. Herz. A universal model for spike-frequency adaptation. *Neural Computation*, 15(11):2523–2564, 2003.

Y. N. Billeh, B. Cai, S. L. Gratiy, et al. Systematic integration of structural and functional data into multi-scale models of mouse primary visual cortex. *Neuron*, 106(3):388–403, 2020. doi: 10.1016/j.neuron.2020.01.040.

G. Chen, F. Scherr, and W. Maass. A data-based large-scale model for primary visual cortex enables brain-like robust and versatile visual processing. *Science Advances*, 8(44):eabq7592, 2022. doi: 10.1126/sciadv.abq7592.

W. Gerstner, M. Lehmann, V. Liakoni, D. Corneil, and J. Brea. Eligibility traces and plasticity on behavioral time scales: experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12:53, 2018. doi: 10.3389/fncir.2018.00053.

N. W. Gouwens, S. A. Sorensen, J. Berg, et al. Classification of electrophysiological and morphological neuron types in the mouse visual cortex. *Nature Neuroscience*, 22:1182–1195, 2019. doi: 10.1038/s41593-019-0417-0.

D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

N. Kriegeskorte, W. K. Simmons, P. S. F. Bellgowan, and C. I. Baker. Circular analysis in systems neuroscience: the dangers of double dipping. *Nature Neuroscience*, 12:535–540, 2009. doi: 10.1038/nn.2303.

T. P. Lillicrap, D. Cownden, D. B. Tweed, and C. J. Akerman. Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7:13276, 2016. doi: 10.1038/ncomms13276.

V. Mante, D. Sussillo, K. V. Shenoy, and W. T. Newsome. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, 503:78–84, 2013. doi: 10.1038/nature12742.

O. Marschall, K. Cho, and C. Savin. A unified framework of online learning algorithms for training recurrent neural networks. *Journal of Machine Learning Research*, 21(135):1–34, 2020.

N. Y. Masse, G. R. Yang, H. F. Song, X.-J. Wang, and D. J. Freedman. Circuit mechanisms for the maintenance and manipulation of information in working memory. *Nature Neuroscience*, 22: 1159–1167, 2019. doi: 10.1038/s41593-019-0414-3.

E. K. Miller, C. A. Erickson, and R. Desimone. Neural mechanisms of visual working memory in prefrontal cortex of the macaque. *Journal of Neuroscience*, 16(16):5154–5167, 1996.

J. M. Murray. Local online learning in recurrent networks with random feedback. *eLife*, 8:e43299, 2019. doi: 10.7554/eLife.43299.

C. K. Pfeffer, M. Xue, M. He, Z. J. Huang, and M. Scanziani. Inhibition of inhibition in visual cortex: the logic of connections between molecularly distinct interneurons. *Nature Neuroscience*, 16:1068–1076, 2013. doi: 10.1038/nn.3446.

H.-J. Pi, B. Hangya, D. Kvitsiani, J. I. Sanders, Z. J. Huang, and A. Kepecs. Cortical interneurons that specialize in disinhibitory control. *Nature*, 503:521–524, 2013. doi: 10.1038/nature12676.

D. Salaj, A. Subramoney, C. Kraisnikovic, G. Bellec, R. Legenstein, and W. Maass. Spike frequency adaptation supports network computations on temporally dispersed information. *eLife*, 10: e65459, 2021. doi: 10.7554/eLife.65459.

C. Teeter, R. Iyer, V. Menon, et al. Generalized leaky integrate-and-fire models classify multiple neuron types. *Nature Communications*, 9:709, 2018. doi: 10.1038/s41467-017-02717-4.

T. Tieleman and G. Hinton. Lecture 6.5 – RMSProp: divide the gradient by a running average of its recent magnitude. COURSERA: Neural Networks for Machine Learning, 2012.

G. G. Turrigiano. The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3): 422–435, 2008. doi: 10.1016/j.cell.2008.10.008.

A. I. Weber and J. W. Pillow. Capturing the dynamical repertoire of single neurons with generalized linear models. *Neural Computation*, 29(12):3260–3289, 2017.

P. J. Werbos. Backpropagation through time: what it does and how to do it. *Proceedings of the IEEE*, 78(10):1550–1560, 1990.

R. J. Williams and D. Zipser. A learning algorithm for continually running fully recurrent neural networks. *Neural Computation*, 1(2):270–280, 1989.

C. Yeung, K. Ganjam, Z. Zhang, S. Mihalas, and Z. Lu. A dataset of differentiable biologically-derived single neuron models. bioRxiv preprint, 2025. URL `https://www.biorxiv.org/content/10.64898/2025.12.02.691895`.

A Supplementary information

A.1 Locked protocol

The confirmatory protocol (`docs/PREREGISTRATION_v11.md`) fixed the cohorts, network, tasks, training budget, learning-rate selection, conditions, endpoints, test families and equivalence margin before any confirmation-cohort run. Its SHA-256, the source-code hash and the analysis-script hash are recorded in `docs/LOCK_v11.json` (Table **??**).

A.2 Connection-probability prior

Table 1: Type-level connection probabilities $P(\text{target} \leftarrow \text{source})$ used to draw the recurrent mask. They are modelling assumptions inspired by Pfeffer et al. (2013); Pi et al. (2013), not measurements for the sampled cells.

target \ source	Pyr	PV	SST	VIP
Pyr	0.20	0.50	0.45	0.05
PV	0.45	0.35	0.30	0.05
SST	0.35	0.05	0.05	0.70
VIP	0.30	0.05	0.40	0.05

A.3 Selected learning rates

A.4 Relation to earlier versions of this project

Earlier internal versions compared intact and shuffled cells while keeping the intact bank's calibration. They split cells by specimen (65 donors were shared between cohorts), compared learners after a fixed short budget at which none had converged, and used one task. Those results are not used here. All numbers in this paper come from the runs described in Methods.